

**RICHFIELD GRADUATE INSTITUTE OF TECHNOLOGY (PTY) LTD  
HIGHER EDUCATION AND TRAINING**

**FACULTY OF INFORMATION TECHNOLOGY**

**BACHELOR OF SCIENCE IT First Year**

**Data Science 500**

**Test 1**

**SEMESTER 2: 2026**

**DATE: 02/09/2026**

**DURATION: 2 HOURS**

**STRUCTURE: QUESTION PAPER**

| NUMBERS                                | QUESTIONS      | MARKS      | TIME IN MINUTES |
|----------------------------------------|----------------|------------|-----------------|
| SECTION A: SHORT QUESTIONS             |                |            |                 |
| 1                                      | Question One   | 20         | 20              |
| SECTION B: LONG QUESTIONS              |                |            |                 |
| 2                                      | Question Two   | 30         | 60              |
| SECTION C: CASE STUDY /ESSAY QUESTIONS |                |            |                 |
| 3                                      | Question Three | 25         | 50              |
| 4                                      | Question Four  | 25         | 50              |
|                                        | <b>TOTAL</b>   | <b>100</b> | <b>180</b>      |

**SECTION A: SHORT QUESTIONS (COMPULSORY)****(20 MARKS)**

- 1.1 Describe the four levels of data (nominal, ordinal, interval, ratio) and give an example of each. **(10 Marks)**
- 1.2 Explain the difference between structured and unstructured data. Provide one example of each. **(5 Marks)**
- 1.3 What is the purpose of exploratory data analysis (EDA) in the data science process? Outline at least three things that a data Scientist will look for during EDA. **(5 Marks)**

**SECTION B: LONG QUESTIONS (COMPULSORY)****(30 MARKS)**

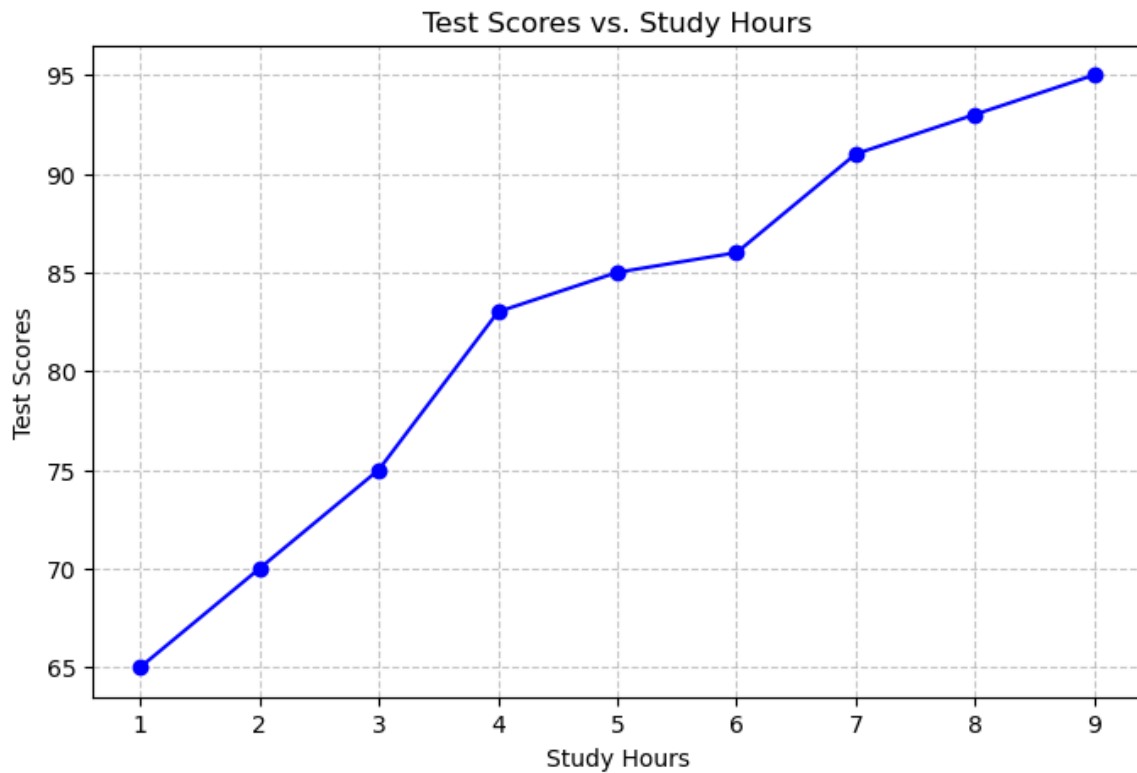
2.1 A certain elective module at Richfield has 20 students. The lecturer is interested in gathering basic information about the performance of these students. To do so, he decides to create a numpy array to store the student's test scores.

```
test_scores = np.array([56, 79, 63, 51, 59, 61, 43, 74, 84, 86, 91, 82, 60, 89, 57, 73, 85, 78, 72, 64])
```

Use your numpy knowledge to answer the following questions:

- a) Write python code that calculates the mean of the test scores. **(3 Marks)**
- b) Write a python statement that calculates the standard deviation. **(3 Marks)**
- c) How would you print out the marks of all students that got a distinction, given that a distinction is a score from 75% to 100%. Write the python code. **(7 Marks)**
- d) What do you understand by the statement:  
"numpy arrays are homogenous" **(2 Marks)**

2.2 In another module the lecturer discovers the following line graph after he gathered information about 9 students.



- a) What can you conclude from the above graph. **(2 Marks)**
- b) Copy and complete the below code that was used to plot the above plot **(13 Marks)**

```
study_hours = [1, 2, 3, 4, 5, 6, 7, 8, 9]
test_scores = [65, 70, 75, 83, 85, 86, 91, 93, 95]

plt.figure(figsize=(8, 5))
plt.____(study_hours, _____, marker='o', linestyle='-',
_____'=blue')

plt.title(_____)
plt.xlabel('Study Hours')
plt.ylabel('Test Scores')

plt.grid(True, linestyle='--', alpha=0.7)

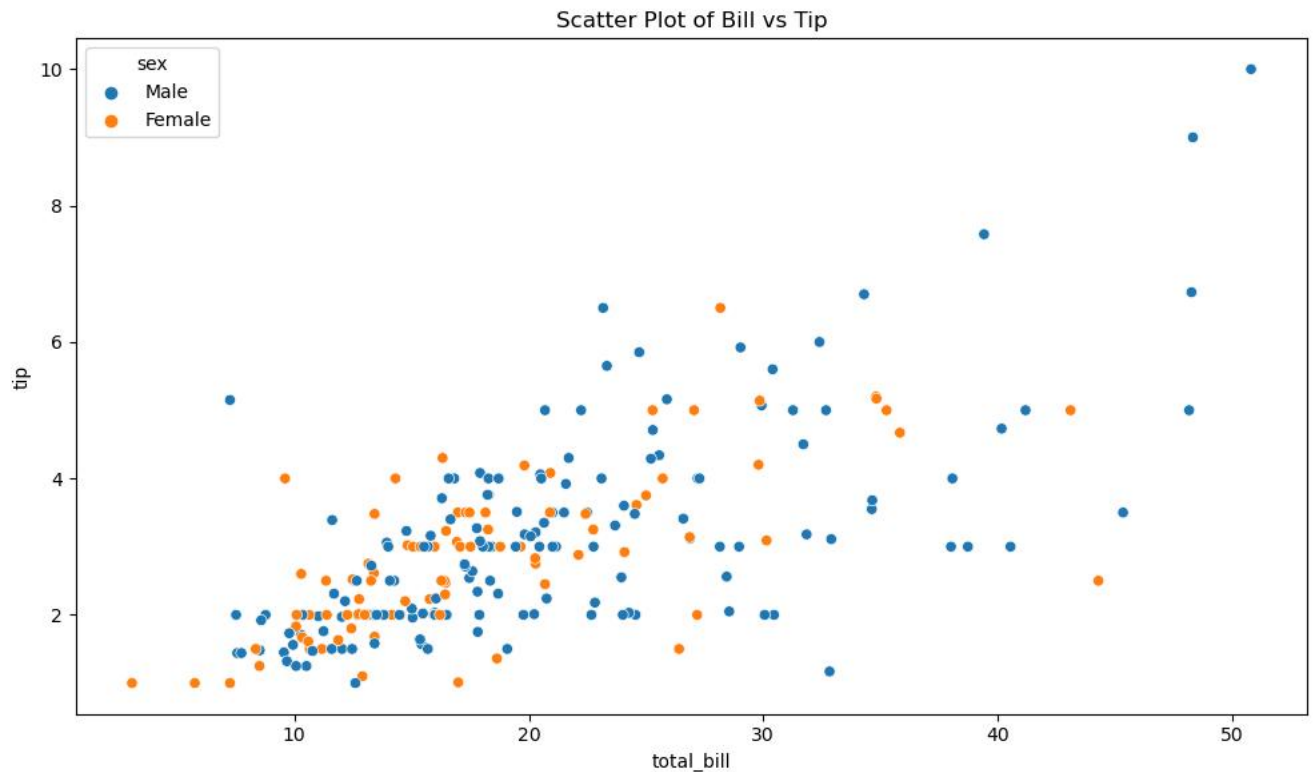
plt.____()
```

**SECTION C: Essay/ Long / Case-Study Questions****(50 MARKS)**

3 The following data is an extract of the tips dataset, this dataset contains information about the tips left by customers at a restaurant.

| <b>total_bill</b> | <b>tip</b> | <b>sex</b> | <b>smoker</b> | <b>day</b> | <b>time</b> | <b>size</b> |
|-------------------|------------|------------|---------------|------------|-------------|-------------|
| 16.99             | 1.01       | Female     | No            | Sun        | Dinner      | 2           |
| 10.34             | 1.66       | Male       | No            | Sun        | Dinner      | 3           |
| 21.01             | 3.50       | Male       | No            | Sun        | Dinner      | 3           |
| 23.68             | 3.31       | Male       | No            | Sun        | Dinner      | 2           |
| 24.59             | 3.61       | Female     | No            | Sun        | Dinner      | 4           |
| 25.29             | 4.71       | Male       | No            | Sun        | Dinner      | 4           |
| 8.77              | 2.00       | Male       | No            | Sun        | Dinner      | 2           |
| 26.88             | 3.12       | Male       | No            | Sun        | Dinner      | 4           |

- a) Write python code that creates the above DataFrame. Hint: (Use a dictionary). **(7 Marks)**
- b) Classify the columns in the DataFrame completely; Each column should be classified as either:  
Quantitative (discrete or continuous)  
Qualitative  
Also specify the level as either; nominal, ordinal, ratio or interval. **(6 Marks)**  
**e.g sex is Qualitative and nominal**
- c) Write a python statement that will give you the statistical summary of all numerical columns in the DataFrame. **(2 Marks)**
- d) The following scatter plot shows the relationship between the total bill and the tip with color coding based on sex of the customers. Provide a convincing explanation of the scatter plot outlining:
- Nature of relationship
  - Gender distribution
  - General conclusion
  - Outliers if any.



4. As a researcher for KGC Data Analytics, you are given the following data from the development of a new test for detecting breast cancer. In clinical trials with 1,000 participants:

- 200 participants actually have breast cancer
- 800 participants do not have breast cancer
- The test correctly identifies 160 people who have breast cancer (true positives)
- The test incorrectly flags 120 people who don't have breast cancer (false positives)

Based on this information, answer the following questions:

- a) What is the probability that a randomly selected participant has breast cancer?  
**(2 Marks)**
- b) What is the sensitivity (true positive rate) of the test? **(5 Marks)**
- c) What is the specificity (true negative rate) of the test? **(5 Marks)**
- d) If a new patient tests positive, what is the probability they actually have breast cancer? (This is the positive predictive value) **(3 Marks)**
- e) Using conditional probability, calculate the probability that a person has breast cancer given that they tested positive (positive predictive value). **(5 Marks)**
- f) Create a confusion matrix to represent this scenario. **(5 Marks)**

**END OF QUESTION PAPER!!!!!!!!!!!!**